

Technical Report: A Dual-Collaborative Knowledge Distillation Framework for Parkinson’s Disease MRI Segmentation

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Abstract. Automatic segmentation of deep gray matter (DGM) nuclei from multimodal MRI is critical for early diagnosis and monitoring of Parkinson’s disease (PD). However, this task is challenging due to low tissue contrast, progressive degenerative changes, and the image resolution. To overcome these challenges, we propose a dual-collaborative knowledge distillation framework with two key innovations: (1) We enable *cross-architecture collaboration* between a convolutional neural network and a Transformer-based foundation model. This design combines local spatial-textural detail with global context modeling enriched by large-scale pretraining knowledge, which enhances fine delineation of small nuclei and improves consistency across heterogeneous modalities. (2) We introduce *hybrid distillation* that transfers confident predictions across models and reinforces stable outputs within each model, which improves robustness and generalization for segmentation task. For the PD-CADxFoundation 2025 multimodal dataset (QSM, NM-MRI, and T1), our method consistently outperforms single-model baselines, with significant Dice improvements, especially on small nuclei (SN $\uparrow$ 16.5%, STN $\uparrow$ 12.9%). These results demonstrate that collaborative knowledge distillation can enhance segmentation accuracy, supporting scalable and clinically meaningful deployment of automated PD diagnosis tools.

Keywords: Parkinson’s disease · Deep gray matter segmentation · Knowledge Distillation · Pseudo-labeling.

1 Introduction

Parkinson’s disease (PD) is a rapidly increasing global health burden, with prevalence projected to reach 25.2 million cases by 2050, largely due to population aging [19]. Current diagnosis still depends on motor symptoms, neurological examination, and response to dopaminergic therapy, commonly evaluated with the Unified Parkinson’s Disease Rating Scale [7]. However, these approaches are subjective, examiner-dependent, and typically detect PD only after significant

neurodegeneration. Magnetic resonance imaging (MRI) provides a non-invasive window into PD pathobiology, where accurate segmentation of deep gray matter (DGM) nuclei is crucial for quantifying biomarkers such as neuromelanin depletion, iron accumulation, and volumetric atrophy, thereby enabling more objective and sensitive tools for early diagnosis and individualized treatment.

Given these diagnostic challenges, the progression of PD is characterized by subtle alterations across multiple DGM nuclei, and no single MRI modality can fully capture the spectrum of pathological changes. Conventional T1-weighted imaging (T1) lacks the resolution to delineate small, low-contrast nuclei such as the substantia nigra (SN) and subthalamic nucleus (STN) [11]. Neuromelanin-sensitive MRI (NM-MRI) provides high specificity for neuromelanin but is largely confined to the substantia nigra pars compacta (SNc) and locus coeruleus [16]. Quantitative susceptibility mapping (QSM) enables sensitive detection of iron deposition; however, its utility is limited by reconstruction variability and lower boundary precision compared with structural MRI [13]. These complementary strengths underscore the necessity of multimodal integration, as recent studies have demonstrated that combining multiple MRI modalities yields richer information for PD characterization [6, 15, 20, 17]. In parallel, deep learning methods have been increasingly applied for PD-specific nuclei segmentation. Approaches based on single or paired modalities [25, 24] can leverage complementary contrasts, yet they still struggle with small nuclei such as the SN, STN, and red nucleus (RN). Multi-sequence strategies have extended segmentation coverage to multiple midbrain nuclei [5], but their performance remains inconsistent across nuclei and subjects. More recently, hybrid CNN–Transformer framework [2] have been explored for PD-related DGM segmentation, but their broader applicability is still constrained by the limited availability of multi-modality datasets.

To address these challenges, we propose a dual-collaborative knowledge distillation framework that couples nnUNet [9] with SwinUNETR [22, 8] as complementary learners. The nnUNet branch provides strong local inductive bias, while the SwinUNETR foundation model brings extensive prior knowledge, global context modeling, and a well-informed initialization that accelerates convergence and stabilizes training. Rather than merging convolution and attention into a single hybrid architecture, we leverage registered multimodal MRI pairs (QSM+T1 or NM+T1) for early voxel-level integration. On the basis, we design a bidirectional distillation strategy: cross-model distillation transfers confident predictions across architectures, while self-distillation reinforces high-confidence voxels within each model. In addition, a feature consistency constraint aligns encoder representations, further improving stability and generalization. Our main contributions are summarized as follows:

- We enable **cross-architecture collaboration** between nnUNet and SwinUNETR, achieving complementary learning that enhances both local detail and global context.
- We introduce a **hybrid distillation scheme** that transfers confident predictions across models and reinforces stable outputs within each model, sup-

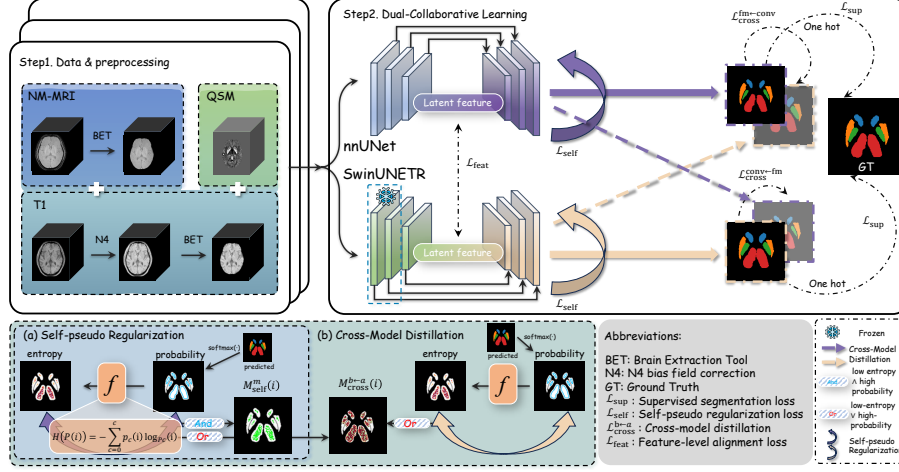


Fig. 1: **Overview of the proposed framework.** **Step 1:** modality-specific preprocessing, followed by normalization, cropping, and augmentation. **Step 2:** dual-collaborative learning with nnUNet (top) and SwinUNETR (bottom), each trained with a structure-aware loss, complemented by self-pseudo regularization, cross-model distillation, and feature alignment losses.

ported by an additional feature consistency constraint for robust representation alignment.

- Our framework achieves **state-of-the-art performance** on multimodal PD nucleus segmentation, with substantial improvements on small and clinically relevant nuclei.

2 Methods

We aim to segment PD-related deep gray matter (DGM) nuclei from *registered* multimodal MRI. Our dual-collaborative framework processes early-fused modality pairs (QSM+T1 or NM+T1) with two complementary learners—a CNN-based nnUNet and a Transformer-based foundation model (SwinUNETR)—and reconciles their outputs via a hybrid distillation scheme with lightweight feature alignment. The pipeline is illustrated in Fig. 1.

2.1 Cross-architecture collaboration

After modality-specific preprocessing, inputs are channel-wise concatenated $X \in \{(QSM, T1), (NM, T1)\}$, with voxel-wise annotations Y provided for the corresponding nuclei, and fed in parallel to nnUNet f_{conv} and SwinUNETR f_{fm} . Each branch produces voxel logits Z^m and probability maps:

$$P^m = \text{softmax}(Z^m), \quad m \in \{\text{conv}, \text{fm}\}. \quad (1)$$

where *conv* indicates the operation of nnUNet and *fm* denotes the operation of SwinUNETR. The nnUNet branch provides strong local inductive bias for fine spatial-textural detail, while the SwinUNETR foundation model supplies global context and rich priors from large-scale pretraining, yielding more stable optimization and faster convergence. Operating the two architectures in parallel on *early-fused*, voxel-aligned inputs preserves their complementary strengths without entangling design constraints of a single hybrid backbone, leading to sharper delineation of small, low-contrast nuclei and improved cross-modality consistency.

2.2 Hybrid distillation scheme

Supervision from ground truth is provided via a structure-aware loss that combines Dice error [4], cross-entropy (CE) [3] and Hausdorff-distance surrogate [10]:

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{Dice}}(P, Y) + \mathcal{L}_{\text{CE}}(P, Y) + w_{\text{HD}}(t) \mathcal{L}_{\text{HD}}(P, Y), \quad (2)$$

where $w_{\text{HD}}(t)$ increases linearly with training epochs, and class weights adaptively emphasize small nuclei.

We further introduce three auxiliary objectives:

(i) *Cross-model distillation* transfers knowledge between branches using a confidence mask

$$M_{\text{cross}}^{b \rightarrow a}(v) = \mathbf{1}(H(P^b(v)) < \tau_H \vee \max(P^b(v)) > \tau_P), \quad (3)$$

where $H(\cdot)$ denotes entropy. Voxels selected by $M_{\text{cross}}^{b \rightarrow a}$ are distilled with temperature-scaled Kullback–Leibler (KL) [12] for large nuclei and CE for small nuclei, yielding $\mathcal{L}_{\text{cross}}$.

(ii) *Self-pseudo regularization* applies a stricter mask

$$M_{\text{self}}^m(v) = \mathbf{1}(H(P^m(v)) < \tau'_H \wedge \max(P^m(v)) > \tau'_P), \quad (4)$$

and enforces CE on these confident pseudo-labels, forming $\mathcal{L}_{\text{self}}$.

(iii) *Feature alignment* encourages consistency between encoder features:

$$\mathcal{L}_{\text{feat}} = \|\phi_{\text{conv}}(X) - \phi_{\text{fm}}(X)\|_2^2. \quad (5)$$

The final objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda(t) \mathcal{L}_{\text{cross}} + \mu(t) \mathcal{L}_{\text{self}} + \gamma(t) \mathcal{L}_{\text{feat}}, \quad (6)$$

where $\lambda(t), \mu(t), \gamma(t)$ gradually increase during training to stabilize learning.

2.3 Implementation Details

Data Preprocessing For method development and ablations, we combined the Stage-1 and Stage-2 labeled cohorts (300 subjects in total) and created an internal stratified split (250 training / 50 validation), ensuring a balanced PD:HC

ratio of 1:1 within each partition. Unless otherwise stated, results are reported on the internal validation set, while final submissions were produced with the model retrained on the full labeled set. The dataset includes T1, QSM, and NM-MRI volumes, which are pre-aligned in each subject’s native space, thus no further inter-modal registration was required. We applied N4 bias-field correction [23] and BET skull stripping [18] to T1 images; NM-MRI volumes were skull-stripped; QSM was used directly as provided by the organizers.

Optimization We trained our framework for segmentation of 18 deep gray matter (DGM) structures using two-channel inputs (T1+QSM or T1+NM), with labels provided on QSM and NM respectively. All models were trained for 500 epochs with a batch size of 2. The CNN branch followed the default nnUNet configuration and was optimized with SGD[21, 1], while the Transformer branch (SwinUNETR) was initialized from pretrained weights [8] (first two layers frozen) and optimized with AdamW[14] (initial learning rate 1×10^{-2} , weight decay 3×10^{-5}). Inputs were z-score normalized within the foreground mask, and cropped into patches of size $56 \times 192 \times 160$ (QSM) or $56 \times 224 \times 160$ (NM). Training and inference were performed on a single NVIDIA A100 80GB GPU (typical memory ~ 10 GB for training, ~ 3 GB for inference).

3 Experiments and Analysis

Unless otherwise specified, all results are reported on the *nnUNet branch*. We denote by “+Self” the nnUNet augmented with self-pseudo regularization, and by “+Self+Cross” the nnUNet trained with the full dual-collaborative scheme (self-pseudo regularization with bidirectional cross pseudo-labeling). SwinUNETR is evaluated as an independent baseline in cross supervision. Feature-level alignment and adaptive structure weighting are consistently enabled across all experiments (including baselines and ablations).

3.1 Evaluation Metrics

We report per-structure: (i) **Dice Similarity Coefficient (DSC)** for overlap quality, and (ii) **95th percentile Hausdorff Distance (HD95)** for boundary discrepancy (in mm). For QSM we additionally summarize *Large*, *Small* and *Overall* averages to reflect scale-aware performance, which are computed over QSM structures only.

3.2 Qualitative analysis

Although SwinUNETR demonstrates competitive performance on large nuclei owing to its ability to capture global context, its Dice scores exhibit higher variance on small and low-contrast structures such as the STN and RN, reflecting instability in boundary delineation. In contrast, nnUNet offers more consistent

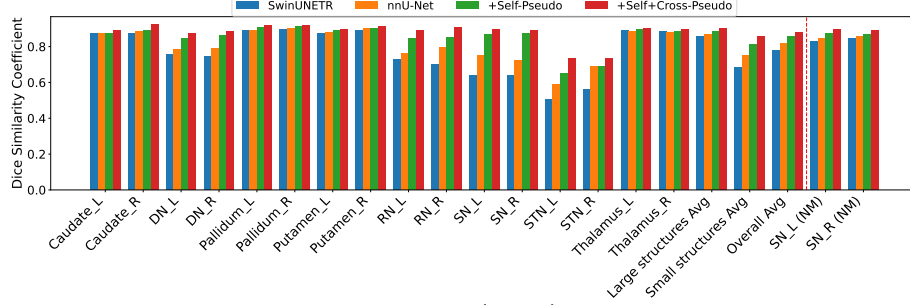


Fig. 2: **Dice Similarity Coefficients (DSC) across structures.** Each bar corresponds to one of the compared strategies: SwinUNETR, nnUNet, +Self-Pseudo, and +Self+Cross-Pseudo. The red dashed line separates QSM results (left, including large/small/overall averages) from NM results (right). The collaborative pseudo-labeling approach consistently improves performance, particularly on small nuclei such as SN, STN, and RN.

segmentation across nuclei while being substantially lighter in parameters and memory footprint, making it more practical for clinical deployment. For this reason, we report quantitative results primarily on the nnUNet branch. Furthermore, compared to a nnUNet baseline trained without foundation model guidance, our dual-collaborative framework achieves a stronger initialization and faster convergence, yielding superior decreasing loss trajectories for both large and small nuclei during training.

3.3 Quantitative Results

Table 1 presents the DSC/HD95 for all structures under four settings (SwinUNETR, nnUNet, +Self, +Self+Cross). Fig. 2 visualizes DSC across structures, with a red dashed divider between QSM (+Averages) and NM.

Baseline performance. nnUNet already performs strongly on large nuclei (e.g., CN/PUT/THA typically DSC > 0.87) but struggles on small and low-contrast structures such as *STN_L* (0.591), *SN_L* (0.751), and *RN_R* (0.793). SwinUNETR is competitive on large structures (e.g., *Pallidum_R* (0.894)) owing to global context, yet its boundary precision on small targets is weaker (e.g., *RN_R* (0.703)).

Effect of self-pseudo. Adding self-pseudo regularization (+Self) consistently boosts DSC, with the most notable gains on small nuclei (e.g., *SN_L*: 0.751 → 0.869, *RN_L*: 0.764 → 0.846). In parallel, HD95 decreases (often approaching 1.0 mm), indicating sharper boundaries. On QSM overall, nnUNet improves from 0.811 to 0.852 DSC (+0.041).

Table 1: Ablation study on QSM and NM-MRI segmentation. Dice similarity coefficient (DSC) and 95% Hausdorff distance (HD95) are reported for each structure under different training strategies. We compare SwinUNETR and nnUNet baselines, nnUNet with self-pseudo labeling (+Self), and nnUNet with both self- and cross-model pseudo-labeling (+Self+Cross). DSC/HD95 values are reported per structure, with higher DSC and lower HD95 indicating better performance.

Structure \ DSC / HD95	SwinUNETR	nnUNet	+Self	+Self +Cross
Caudate_L	0.874 / 1.45	0.872 / 1.40	0.876 / 1.35	0.893 / 1.34
Caudate_R	0.871 / 1.22	0.883 / 1.18	0.893 / 1.15	0.921 / 1.14
DN_L	0.755 / 1.30	0.785 / 1.27	0.846 / 1.24	0.873 / 1.22
DN_R	0.747 / 1.33	0.791 / 1.31	0.860 / 1.29	0.884 / 1.28
Pallidum_L	0.892 / 1.05	0.888 / 1.03	0.907 / 1.01	0.919 / 1.00
Pallidum_R	0.894 / 1.05	0.901 / 1.02	0.913 / 1.01	0.919 / 1.00
Putamen_L	0.874 / 1.12	0.879 / 1.10	0.892 / 1.08	0.896 / 1.04
Putamen_R	0.890 / 1.20	0.901 / 1.18	0.902 / 1.16	0.915 / 1.16
RN_L	0.728 / 1.10	0.764 / 1.08	0.846 / 1.04	0.893 / 1.00
RN_R	0.703 / 1.12	0.793 / 1.10	0.849 / 1.05	0.909 / 1.00
SN_L	0.641 / 1.20	0.751 / 1.18	0.869 / 1.04	0.896 / 1.00
SN_R	0.638 / 1.22	0.721 / 1.20	0.871 / 1.06	0.893 / 1.00
STN_L	0.503 / 1.55	0.591 / 1.52	0.649 / 1.50	0.734 / 1.47
STN_R	0.563 / 1.40	0.687 / 1.37	0.689 / 1.35	0.734 / 1.33
Thalamus_L	0.891 / 1.08	0.887 / 1.06	0.894 / 1.05	0.902 / 1.04
Thalamus_R	0.885 / 1.31	0.879 / 1.30	0.883 / 1.29	0.894 / 1.28
SN_L (NM)	0.829 / 1.08	0.843 / 1.06	0.872 / 1.03	0.896 / 1.00
SN_R (NM)	0.847 / 1.08	0.856 / 1.06	0.866 / 1.03	0.893 / 1.00

Effect of cross-pseudo. With cross pseudo-labeling (+Self+Cross), nnUNet further improves by leveraging complementary cues from SwinUNETR. The structure-aware hybrid supervision (KL for large, CE for small) yields additional gains, especially on difficult nuclei (e.g., *STN_R*: 0.687→0.734, *SN_R (NM)*: 0.856→0.893) while keeping HD95 close to 1.0 mm on most classes. Overall QSM DSC rises to 0.880 (+0.069 vs. baseline), with small-set mean improving from 0.718 to 0.843 (+0.125).

4 Limitations and Future Work

Despite the performance gains of our dual-collaborative framework, several limitations remain. First, small and low-contrast nuclei such as STN still show sub-optimal Dice scores under weak contrast or partial volume effects, suggesting the need for anatomical priors to constrain spatial localization or uncertainty modeling to down-weight unreliable pseudo-labels and prevent error propagation. Second, the effectiveness of pseudo-labeling is influenced by the quality of early predictions, especially those from the foundation model. Although confidence filtering reduces noise, false positives may still propagate, particularly for rare

structures. Third, our current experiments are restricted to a fully supervised setting. In practice, large amounts of unlabeled MRI data are available, and incorporating them through semi-supervised or self-supervised learning could improve sensitivity to small nuclei and stabilize boundary delineation. Future work should evaluate the framework across external cohorts to assess clinical transferability. In addition, exploring end-to-end integration may more effectively leverage multimodal complementarity.

Looking ahead, we envision incorporating this paradigm into real-time clinical pipelines, potentially combined with interactive AI tools that allow expert users to guide or correct model outputs. Extending the framework to multi-institutional datasets and diverse acquisition protocols will also be essential to ensure generalizability, fairness, and clinical applicability in Parkinson’s disease diagnosis and monitoring.

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